The Product Rule: The Product Rule states that:

$$\frac{d}{dx}\left[u(x)v(x)\right] =$$

Thus, we can apply the Fundamental Theorem of Calculus and obtain

$$u(x)v(x) = \int \qquad\qquad + \int \qquad\qquad .$$

Rearranging the terms, we get the

Integration by Parts Formula:

How do we pick u and dv? Well, to start with, dv should to be easy to integrate!
Examples:

1. $\int xe^x\,dx$

   $u =$ $dv =$

   $du =$ $v =$

2. $\int_1^2 \ln x\,dx$

   $u =$ $dv =$

   $du =$ $v =$

3. $\int x \ln x \, dx$

4. $\int_0^{\pi} x \sin x \, dx$

5. $\int_0^1 x\sqrt{x+1} \, dx$

6. $\int x^2 e^x \, dx$ (Hint: Integrate by parts twice)

7. $\int \arctan x \, dx$ (Hint: Let $u = \arctan x$.)

8. $\int e^x \cos x \, dx$ (Hint: Integrate by parts twice.)

9. $\int x(\ln x)^3 \, dx$ (Hint: Integrate by parts three times.)

10. $\int xe^{x^2} \, dx$ (Hint: Don't work too hard!)